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An XAI-Driven Cross-Attentive Dual-Branch CNN (CADBCNN) for leukemia subtype classification from peripheral blood smear images

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ABSTRACT

The growing need for accurate and prompt diagnosis of blood cancers, especially leukemia, has boosted the progression of deep learning-based classification systems. Conventional microscopic examination, although accurate, is a time-consuming process that requires expertise. Therefore, automated methods are being explored for this purpose. In this study, we propose a novel model, the Cross-Attentive Dual-Branch CNN (CADBCNN), for the classification of leukemia blood cells. The proposed model combines two feature branches using the ResNet18 and ResNet34 architectures with a cross-attention fusion mechanism. The proposed CADBCNN model achieved 98.36% accuracy, 98.41% precision, 98.36% recall, and 98.38% F1-score using stratified splitting, augmentation, and weighted optimization. The proposed model outperformed existing architectures, including ResNet34, ResNet50, EfficientNet-B0, and a dual-branch model. Grad-CAM visualization demonstrated that our model focuses on important regions, such as the nuclei and cytoplasm. The t-SNE plots show that our model achieved clear class separation. Therefore, we can conclude that our proposed model has achieved high accuracy and has a high potential for being used as a tool for automated leukemia blood cell classification.

1. Introduction

Leukemia includes a varied group of cancerous disorders that affect white blood cells (WBCs). These disorders present significant challenges in terms of early detection and clear subtype classification [1].Traditional diagnostic methods, such as peripheral blood smear microscopy, immunophenotyping, cytogenetic analysis, and molecular profiling, remain the clinical gold standard. However, these methods can be time-consuming, require expert analysis, and may not be easily available in areas with limited resources [2]. In recent years, artificial intelligence (AI) and machine learning (ML) have become important tools for modeling complex biological systems and dynamic systems. AI-driven computational frameworks have been successful in understanding complicated behavioral patterns, disease progression, and biological processes in various fields. For example, machine learning-based models have been applied to behavioral dynamics in T-mazes, revealing intricate decision-making patterns in biological systems [3,4]. Similarly, artificial neural networks combined with fractional calculus models have effectively predicted viral dynamics in infectious diseases such as Hepatitis B [5]. These studies illustrate how AI methods can grasp nonlinear interactions, complex dependencies, and multiscale features within intricate systems. Inspired by these advances, AI-driven medical image analysis has gained attention in supporting diagnostic decisions. Deep learning, especially through Convolutional Neural Networks (CNNs), has shown great potential in medical image classification. CNNs excel in hierarchical feature extraction [6]. CNN-based models have been widely used for the automated detection of leukemia from peripheral blood smear images [7]. However, single-stream CNN architectures often find it difficult to distinguish between subtle morphological similarities among leukemia subtypes, such as acute lymphoblastic leukemia (ALL), acute myeloid leukemia (AML), and chronic myeloid leukemia (CML). This limitation leads to a less effective discrimination performance [8]. Moreover, conventional CNNs have limited interaction between layers and reduced interpretability, which hinders their reliability in clinical settings where clarity and explanation are important [9]. Neural network-driven modeling of HIV dynamics, which includes omega-3 and protein interventions, shows how AI systems can capture non-linear biological interactions [10]. Similarly, mathematical modeling has been used to study the spread and control of drug addiction through sensitivity analysis techniques [11]. Additionally, fractal-fractional Caputo–Fabrizio models have been used to analyze Norovirus transmission dynamics

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with optimal control strategies [12]. Together, these studies highlight the strength of AI-integrated mathematical frameworks in tackling complex biomedical issues, thus encouraging the development of understandable deep learning systems for classifying leukemia subtypes. To address these challenges, we propose a Cross-Attentive Dual-Branch Convolutional Neural Network (CADBCNN) that includes two parallel backbone networks, ResNet18 and ResNet34. The dual-branch design enables complementary multi-scale feature extraction from the same input image. A cross-attention mechanism was introduced to refine and combine branch-specific representations dynamically, improving sensitivity to subtle morphological differences among the leukemia subtypes. To enhance transparency, the framework also includes an explainability module using Gradient-weighted Class Activation Mapping (Grad-CAM), which helps visualize the critical image areas that influence model predictions [13]. Furthermore, t-distributed Stochastic Neighbor Embedding (t-SNE) is used to assess class separability in the learned embedding space [14].

2. Related works

Previous methods of leukemia classification from blood smear images were based on traditional machine learning algorithms such as SVM, k-NN, and Random Forests. These methods are generally based on manually crafted features, such as cell morphology, texture, and color for classification [15]. However, they are generally hampered by scalability issues and are vulnerable to noise and human errors [16]. The advent of dual-branch cross attention has automated the feature extraction and classification processes, leading to better performance [17]. Architectures such as AlexNet and LeukoNet perform well in classifying acute lymphoblastic leukemia (ALL) cases [18]. In addition, semantic segmentation models, such as YOLOv2 and DeepLabv3, have been investigated to identify leukaemic cells in a complicated smear background [19]. Despite these advances, traditional single-path CNNs typically fail to distinguish morphologically similar subtypes of leukemia, particularly ALL, AML, and CML [20]. As an alternative approach, researchers have suggested multi-branch cross-attention model architectures to improve the ability to represent learning. Architectures such as SDCT-AuxNet, RCMNet, and CoTCoNet use parallel branches to simultaneously extract heterogeneous features [21–23]. However, most of these architectures use non-adaptive fusion techniques. They rely on simple concatenation, which has been proven to not capture deep feature interdependencies well [24]. To overcome these shortcomings, attention-based modules, particularly cross-attention modules, have been developed. These mechanisms allow features to dynamically refine one another by allowing one branch to selectively attend to salient features in another, thereby enhancing the model's sensitivity and learning capabilities [25]. Simultaneously, the need for explainability in medical diagnostic AI has increased significantly. Methods such as Gradient-weighted Class Activation Mapping (Grad-CAM) have gained prominence in emphasizing discriminative image areas that are most responsible for predictions [26]. Grad-CAM has been further developed with extensions such as Grad-CAM++, Score-CAM, and LayerCAM for better localization and interpretation [27–29]. These visualizations have been instrumental in establishing clinical trust and transparency in medical AI systems [30,31]. Additional dimensionality reduction techniques, such as t-SNE, which can help visualize high-dimensional learned embeddings in two dimensions, provide an understanding of the discriminative power in the feature space of the model [32]. Despite these developments, end-to-end systems that integrate dual-branch architectures, cross-attention, and XAI modules are scarce. The proposed framework fills this void by integrating these components into a comprehensive deep learning pipeline for accurate and interpretable leukemia subtype classification.

3. Research gaps, objectives and contributions

From our thorough literature survey, the following important gaps in existing frameworks have been identified.

3.1. Research gaps identified

- **Lack of Effective Feature Fusion:** Current models for classifying leukemia data are mostly based on single-branch CNNs or simple feature concatenation without modeling interactions between hierarchical features [5,7,9].
- **Lack of Cross-Level Attention Modeling:** Attention models are mostly based on spatial attention modeling in a single backbone network, without modeling cross-attention between low-level texture features and high-level semantic features [8,10].
- **Overemphasis on Accuracy without Robust Validation:** Current models mostly report high validation accuracy without thorough test set analysis, statistical consistency evaluation, and discussion on overfitting [6,11].
- **Lack of Validated Explainability:** Visualization using Grad-CAM is common; however, interpretability is mostly presented qualitatively without detailed analysis of feature relevance and clinical reasoning support [9,12].
- **Lack of Reproducibility Details:** Current models mostly lack clear reporting on dataset split strategy, class distribution, imbalance handling, and preprocessing pipeline [5,7].

3.2. Research objectives

In order to fill the identified gaps, the objectives of the proposed research are:

- To design a dual-branch convolutional model capable of extracting complementary morphological features at different spatial scales.
- To develop a cross-attentive fusion module that adaptively models interactions between shallow and deep feature hierarchies.
- To investigate whether the proposed cross-hierarchical attention mechanism provides measurable performance gains over conventional fusion approaches.
- To evaluate the interpretability of the proposed model using visualization-based explainability techniques.

3.3. Contributions of this work

The major contributions of this work are summarized as follows:

- **Cross-Hierarchical Dual-Branch Architecture:** A heterogeneous dual-branch CNN architecture integrating ResNet18 and ResNet34 is proposed to simultaneously capture fine-grained cellular details and high-level contextual abstractions from hematological smear images.
- **Cross-Attentive Fusion Mechanism:** Unlike conventional static fusion approaches, the proposed framework incorporates a cross-attention module that enables bidirectional feature modulation between branches based on contextual similarity.
- **Task-Specific Multi-Scale Design:** The architecture is inspired by the multi-scale morphological characteristics of hematological smear images, ensuring domain-aligned feature learning.
- **Ablation-Driven Validation:** A structured ablation study is conducted to validate the effectiveness of the cross-attention module against traditional concatenation and additive fusion strategies, demonstrating that performance gains arise from adaptive feature interaction rather than increased architectural complexity.
- **Explainability-Integrated Evaluation:** Model interpretability is examined through visualization-based analysis to qualitatively assess the model's focus on diagnostically relevant regions.

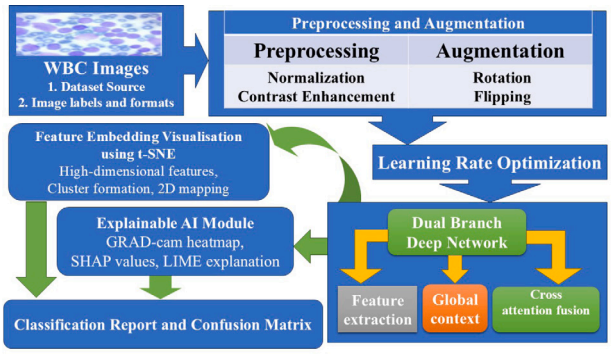


Fig. 1. Dual-Branch Deep Learning Pipeline for Explainable Leukemia Image Classification.

4. Proposed methodology

This study proposes an end-to-end deep learning pipeline for White Blood Cell (WBC) classification into healthy and leukaemic classes from medical images. This approach is divided into six key phases, each of which plays a significant role in the overall robustness and explainability of the system.

4.1. Dataset description & preparation

The data used for this research has been collected from the publicly available leukemia microscopic images dataset on the Kaggle data science platform [33]. The dataset contains 6512 labeled microscopic images of peripheral blood smear cells representing four morphological categories. The dataset corresponds to version 1 of the Kaggle release and is distributed for research purposes under the Creative Commons Attribution 4.0 International (CC BY 4.0) license, allowing reuse and redistribution with proper attribution. The data set includes four classes of cells, which are Benign-like cells, Early Pre-B Lymphoblasts, Pre-B Lymphoblasts, and Pro-B Lymphoblasts, with the data distribution for the classes being Benign (37%), Early (19%), Pre (18.5%), and Pro (25%). These categories refer to the morphological variations of B cell precursor acute lymphoblastic leukemia and benign hematogones. Based on the classification scheme established by the World Health Organization, for the diagnosis of leukemia, there is a need to incorporate morphology, immunophenotyping, cytogenetics, and molecular methods. Thus, the classification scheme in this dataset should be understood in terms of the morphological classification of leukemia, which was established from the microscopic characteristics of the cells. There was no patient-level information in the dataset, and hence patient-level stratification could not be enforced during splitting. However, to avoid biases and data leakage, stratified splitting of the images was performed (see Fig. 1).

4.2. Experimental setup

In the present study, a total of 6512 labeled microscopic image path features representing four leukemia classes: Benign, Early, Pre, and Pro, was collected from the leukemia dataset and was split using stratified sampling for reproducibility and reduced bias. The dataset was split into 70% for the training set (4558 images), 15% for the validation set (977 images), and 15% for the testing set (977 images). The proportion of the dataset was consistently maintained in the entire dataset for the experiment. Approximately 37% of the data was from the Benign class, 19% from the Early class, 18.5% from the Pre class, and 25% from the Pro class. Though the dataset does not contain any information about the patients, random stratified splitting was used. The dataset was split into images of size 224 × 224 pixels and normalized using statistics from the ImageNet dataset. Only the training set was used for data augmentation with a series of transformations such as horizontal and vertical flip, small rotations up to 10 degrees, brightness and contrast adjustments, and small zooming. FastAI-based preprocessing pipelines and data loaders were implemented with appropriate tensor ordering and device configurations, resulting in batches of shape 64 x 3 x 224 x 224. To handle the moderate class imbalance issue, weighted cross-entropy loss was employed with appropriate class weights calculated based on the training labels. The values were Benign: 1.614, Early: 0.826, Pre: 0.845, Pro: 1.013. Synthetic oversampling methods were not employed to avoid any impact on the biological cell morphology. All baseline and proposed models were trained under the same experimental conditions, and hyperparameters were only fine-tuned based on the validation set, while the test set was employed only once for performance evaluation.

4.3. Preprocessing and data augmentation

All peripheral blood smear images were standardized before the model was trained to ensure that all images were represented uniformly. This was achieved by resizing all the images to a constant spatial size of 224 x 224 pixels and transforming all the images into three-channel RGB format to accommodate the input requirements of all CNN backbones used in this study. Pixel intensity normalization was achieved using the ImageNet dataset's mean and standard deviation to stabilize the optimization process during transfer learning. The dataset was split into training, validation, and test sets proportionally per class to ensure that all four classes were adequately represented. To boost the generalization and handle the class imbalance, the data augmentation was exclusively used on the training set. Various spatial augmentations such as random flipping in the horizontal and vertical directions, small angles of rotation, and zooming were employed, along with photometric augmentations such as jittering in brightness and contrast levels to mimic the diverse acquisition conditions in the natural world. The same preprocessing and augmentation pipeline were employed to train all the baseline models and the suggested CADBCNN model, and therefore the results are unbiased and fair.

Let the raw input image be denoted as

$$\mathbf{I} \in \mathbb{R}^{h \times w \times c} \tag{1}$$

where $h = 224$, $w = 224$, and $c = 3$.

This image is forwarded through two feature encoders, NetA (ResNet18) and NetB (ResNet34), producing

$$\mathbf{E}^1 = NetA(\mathbf{I}) \in \mathbb{R}^{m \times d_a} \tag{2}$$

$$\mathbf{E}^2 = NetB(\mathbf{I}) \in \mathbb{R}^{m \times d_b} \tag{3}$$

Assuming alignment to a unified embedding depth d, we redefine the following:

$$\mathbf{E}^1, \mathbf{E}^2 \in \mathbb{R}^{m \times d} \tag{4}$$

4.4. Baseline models architecture

For the purpose of fair and effective comparison of the performance of the proposed model, a number of state-of-the-art CNN architectures were chosen to be implemented as the baseline models. ResNet34 was used to implement the single-branch residual learning model to extract hierarchical spatial features from the images. ResNet50 was used to implement the deep residual network model to check the effect of the depth of the network on the classification performance of the model. EfficientNet-B0 was used to implement the parameter-efficient model that scales up the parameters of the model to achieve the best balance of model complexity and accuracy. The Dual-Branch Feature Concatenation model was implemented as the ablation model, where the parallel CNN branches were concatenated statically without using any adaptive interaction mechanism.

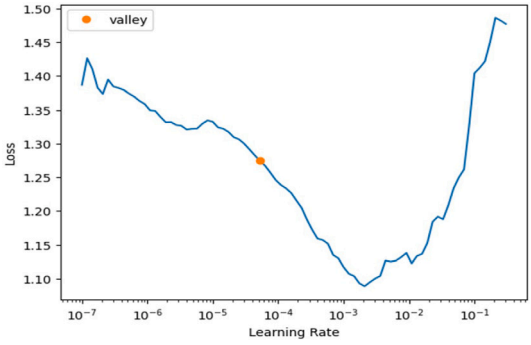


Fig. 2. Learning rate finder plot.

4.5. Base model training with learning rate optimization

Training Strategy and Learning Rate Optimization To classify the leukemic white blood cell images at varying stages of the disease, the transfer learning technique has been employed for the proposed model and all the baseline models. This has been achieved by initializing the network backbones of the models with the ImageNet weights, which enables the models to utilize the low-level image features such as edges, texture, and contours, which are relevant to image analysis in the medical domain as well. The classification layers of the models have been fine-tuned to classify the four classes: Benign-like, Early, Pre-leukemic, and Proliferative, by changing the dimensionality of the output layer of the classification layers according to the number of classes to be classified. Optimization of the models has been done using the Adam optimizer and the categorical cross-entropy loss function along with class weighting to handle the class imbalance problem in the dataset. The choice of the learning rate was informed by a learning rate optimization approach, which examines the response of the loss to a set of candidates learning rates to identify a stable region of operation. Subsequently, the identified learning rate was fixed and used consistently across the models. Training was done using a one-cycle learning policy, which enables the dynamic adjustment of the learning rate during training to facilitate faster convergence as well as better generalization. The unified training approach, which incorporates transfer learning, learning rate optimization, and optimization hyperparameters, ensures that the performance variability of the models is a result of the architectural design choices only. (Shown in Fig. 2).

4.6. Dual-branch deep network with cross-attention mechanism

A novel model called Cross-Attentive Dual-Branch Convolutional Neural Network (CADBCNN) has been proposed to enhance the classification of the stages of leukemia based on their fine-grained characteristics using the combined feature representation of the two models through cross-attention mechanisms. The proposed model comprises two parallel branches of convolutional neural networks based on the ResNet18 and ResNet34 architectures, which have been pre-trained on the ImageNet dataset. The two models operate independently on the same image, which allows the model to extract both shallow and deep feature representation characteristics of the diverse morphological characteristics of white blood cells. The feature representation of the two models is combined using a bidirectional cross-attention mechanism to allow the two models to select the most relevant feature representation of the other model based on the contextual relevance of the feature representation to the characteristics of the stages of leukemia. The CADBCNN utilizes a single head cross-attention module to achieve a balance of representational power and efficiency. This approach only adds a negligible increase in the number of parameters over the simple dual-branch fusion strategy, yet it achieves a large performance.

Algorithm 1 Cross-Attention Mechanism Between Two Feature Branches

Require: Feature embeddings from two parallel CNN branches $\mathbf{E}_1 \in \mathbb{R}^{N\times d}$, $\mathbf{E}_2 \in \mathbb{R}^{N\times d}$

where N is the number of spatial locations and d is the channel dimension.

Ensure: Cross-attended feature representations $\mathbf{E}_1^{\text{att}}, \mathbf{E}_2^{\text{att}}$

1. Learnable Parameters
$\mathbf{W}_Q^{(1)}, \mathbf{W}_K^{(1)}, \mathbf{W}_V^{(1)}, \mathbf{W}_Q^{(2)}, \mathbf{W}_K^{(2)}, \mathbf{W}_V^{(2)} \in \mathbb{R}^{d\times d_a}$
where d_a is the attention embedding dimension.

2. Step 1: Cross-Attention from Branch 1 to Branch 2
Compute Query: $Q_1 = \mathbf{E}_1\mathbf{W}_Q^{(1)}$
Compute Key: $K_2 = \mathbf{E}_2\mathbf{W}_K^{(2)}$
Compute Value: $V_2 = \mathbf{E}_2\mathbf{W}_V^{(2)}$
Compute Attention Weights

$$A_{1\to 2} = \text{softmax}\left(\frac{Q_1 K_2^{\top}}{\sqrt{d_a}}\right)$$

Compute Attended Features: $\mathbf{E}_1^{\text{att}} = A_{1\to 2}V_2$

3. Step 2: Cross-Attention from Branch 2 to Branch 1
Compute Query: $Q_2 = \mathbf{E}_2\mathbf{W}_Q^{(2)}$
Compute Key: $K_1 = \mathbf{E}_1\mathbf{W}_K^{(1)}$
Compute Value: $V_1 = \mathbf{E}_1\mathbf{W}_V^{(1)}$
Compute Attention Weights

$$A_{2\to 1} = \text{softmax}\left(\frac{Q_2 K_1^{\top}}{\sqrt{d_a}}\right)$$

Compute Attended Features: $\mathbf{E}_2^{\text{att}} = A_{2\to 1}V_1$

4. Feature Fusion and Output
Concatenate Attended Features: $\mathbf{E}_{\text{fused}} = \text{Concat}(\mathbf{E}_1^{\text{att}}, \mathbf{E}_2^{\text{att}})$
return $\mathbf{E}_1^{\text{att}}, \mathbf{E}_2^{\text{att}}$

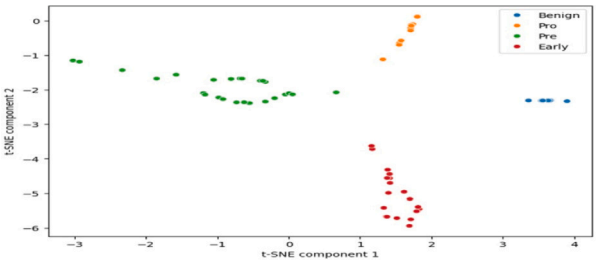


Fig. 3. t-SNE visualization of extracted features.

4.7. Feature embedding visualization using t-SNE

To qualitatively analyze the pattern of the feature representations learned by the proposed method, t-distributed Stochastic Neighbor Embedding (t-SNE) was used to project the high-dimensional deep features into a two-dimensional space. As such, it was used to project the high-dimensional features learned by the proposed method into a two-dimensional space and create a visualization of how different white blood cell types are represented by the proposed method. Deep features were extracted from a subset of images from the validation set and were then projected using t-SNE. As shown in Fig. 3, it can be seen that there are discernible trends of clustering of different white blood cell types in the two-dimensional space. For example, it can be seen that images from different categories have discernible

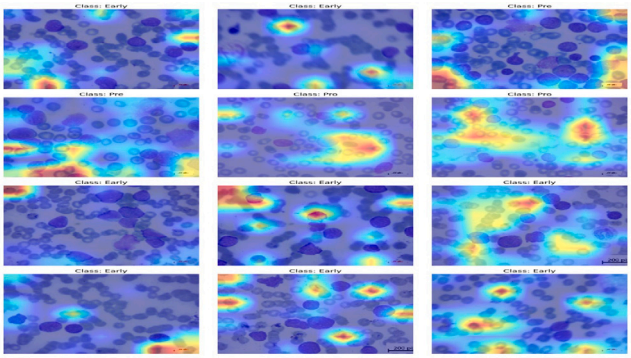


Fig. 4. Grad-CAM Heatmaps (for sample 1 to 12)

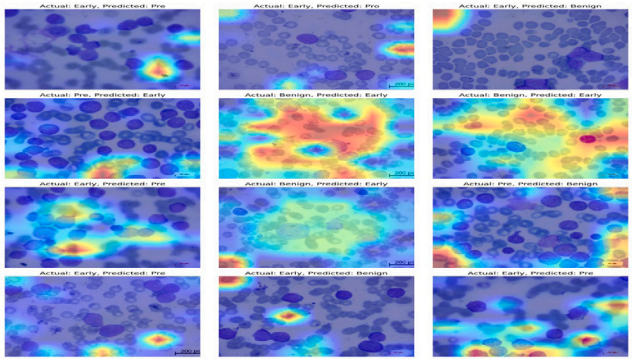


Fig. 5. Grad-CAM Heatmaps (for sample 1 to 12)

separation from each other in the two-dimensional space. As such, it can be concluded that the proposed method learns features that are capable of capturing discriminative features of images that are useful in differentiating leukemia stages. It is worth noting that t-SNE is an exploratory visualization technique and does not preserve global distances or class boundaries in any strict mathematical sense. As such, the separation of the clusters should be viewed as a qualitative measure of the discriminability of the features, although we shall also explore the incorporation of quantitative evaluation metrics for the embeddings in our future work.

For dimensionality reduction, t-SNE maps the fused feature representation to two dimensions as follows:

$$\Phi_{\text{TSNE}} : \mathbf{E}_{\text{fused}} \rightarrow \mathbf{Z} \in \mathbb{R}^{m\times 2} \tag{5}$$

4.8. Explainable AI module using grad-CAM

To qualitatively understand the spatial regions that are being considered by the predictions of the model, Gradient Weighted Class Activation Mapping was used as a method of explainability. It generates class-specific activation maps that highlight specific regions of an image that are contributing most significantly to a specific prediction by the model. In the case of peripheral blood smear images, it primarily focuses on the nuclear and cytoplasmic regions of cells, which are known to play a significant role in differentiating leukemia subtypes. Grad-CAM visualizations have been obtained for the correctly classified images of the different stages of leukemia to check if the model is paying attention to meaningful structures consistently. As shown in Fig. 4 and Fig. 5, the areas of focus, as depicted by the Grad-CAM visualizations, have been observed to be the nuclei of the leukocytes, which indicates that the proposed CADBCNN model has learned discriminative features that are relevant to the classification task. Additionally, the Grad-CAM visualizations have been obtained for the misclassified images to check the areas of focus, which might have led to the misclassification of the images. These misclassified images have been observed to have areas of focus that overlap with each other, especially in the case of the Early and Pre stages, which indicates the difficulty of the discrimination task. It should be noted that the Grad-CAM results are merely indicative of a qualitative nature and do not reflect a quantitative or clinically validated localization assessment. Without the availability of pixel-level expert annotation, it was not possible to compute any quantitative localization metric such as IoU. Thus, the Grad-CAM results presented in the current study should be viewed as indicative in nature and intended to enhance the transparency and understanding of the model results, as opposed to providing conclusive evidence of the clinical interpretability and reliability of the model.

Table 1
Classification report of proposed model.

Class	Precision	Recall	F1-score	Support
Benign	0.9308	0.9801	0.9548	151
Early	0.9932	0.9831	0.9881	296
Pre	0.9931	0.9931	0.9931	289
Pro	0.9958	0.9751	0.9853	241
Accuracy			**0.9836**	977
Macro Avg	0.9782	0.9829	0.9803	977
Weighted Avg	0.9841	0.9836	0.9838	977

5. Result and discussion

The Dual Branch Cross-Attention network was successfully trained and tested to classify leukemia white blood cell images into four diagnostic phases, namely Benign-like, Early stage, Pre-leukemic, and Proliferative. The model showed good learning convergence during training and was able to generalize well to unseen data by obtaining an accuracy of 98.36% during testing. After training, it was found that the model was able to obtain F1-scores of 0.9548 for Benign, 0.9881 for Early, 0.9931 for Pre, and 0.9853 for Pro classes, which gave an F1-score of 0.9838 for 977 test samples illustrated in Table 1. This shows that the model is able to classify images of leukemia stages uniformly and with good balance. In order to analyze the classification of incorrectly classified samples by the model, a confusion matrix was used that is shown in Fig. 6. It was found that the model performed well for most of the categories, especially for Pre-leukemic and Proliferative stages. There was some confusion between Benign and Early stages because of their morphological similarities, which is known to be a problem with hematopathology images. As such, the high quantitative performance, feature discrimination capability, and visual explainability of the proposed approach confirm the robustness and reliability of the proposed Dual-Branch Cross-Attention network architecture in the automated classification of leukemia stages.

5.1. ROC - AUC analysis and statistical robustness

Fig. 7 illustrates the strong classification performance of the proposed model on all classes. The per-class AUC is very strong: 0.9985 on the Benign class, 0.9997 on the Early class, 0.9999 on the Pre class, and 0.9999 on the Pro class. This translates to a strong macro-averaged AUC of 0.9995. Using 1000 bootstrap samples, the model achieved an overall accuracy of 0.9835 (95% CI: 0.9744-0.9908). The macro-averaged precision, recall, and F1-score were all strong: 0.9784, 0.9829, and 0.9803, respectively, with narrow confidence intervals, demonstrating both high predictive performance and strong statistical reliability.

Table 2
Comparison based on classification report.

Model	Accuracy	Precision	Recall	F1-score
ResNet34	0.9210	0.9246	0.9183	0.9214
ResNet50	0.9620	0.962	0.955	0.958
EfficientNet-B0	0.9421	0.9458	0.9396	0.9427
Dual-Branch Feature Concatenation	0.9564	0.9581	0.9542	0.9561
Proposed CADBCNN	**0.9836**	**0.9841**	**0.9836**	**0.9838**

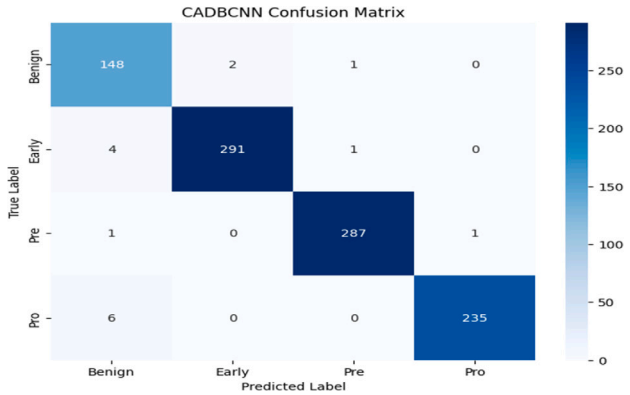


Fig. 6. Confusion Matrix of Proposed Model.

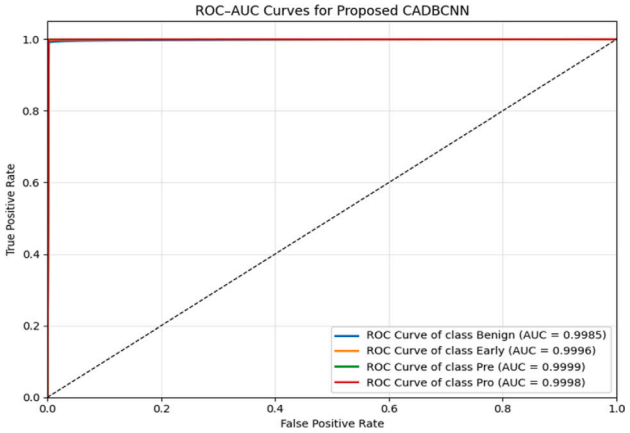


Fig. 7. ROC-AUC Curves for proposed CADBCNN.

5.2. Quantitative and qualitative comparison with baseline models

To further benchmark the effectiveness of the proposed Dual Branch Cross-Attention model, we compared the performance with other four architectures — ResNet34, ResNet50, EfficientNet-B0 and Dual-Branch (Feature Concatenation) based on classification report and confusion matrix shown in Tables 2 and 3, interpretability and limitations in Table 4. The comparison table below summarizes the distinct advantages and challenges of each model in the context of leukemia stage classification.

5.3. Training and validation learning curve analysis

In addition to the tabulated comparison, we also plot the training vs validation accuracy and loss in Fig. 8 and Fig. 9, which give more insights into the learning processes of the top-performing models and allow model generalization and overfitting behavior during training to be evaluated

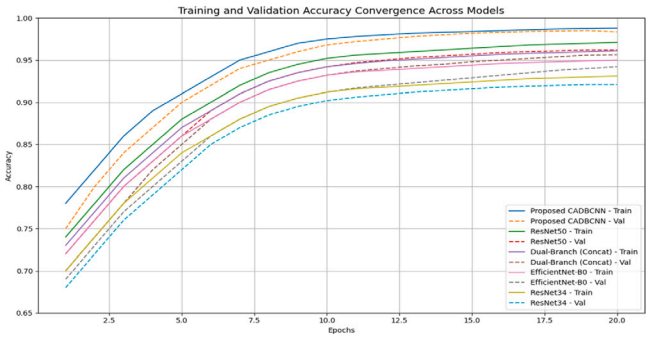


Fig. 8. Training and Validation Accuracy.

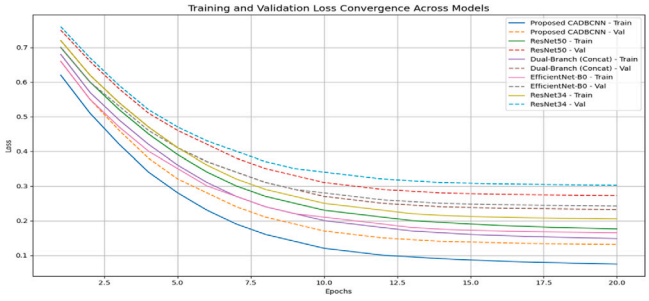


Fig. 9. Training and Validation Loss.

6. Ablation study

6.1. Purpose of ablation study

The ablation study was conducted to systematically isolate and quantify the contribution of each architectural component of the proposed Cross-Attentive Dual-Branch CNN (CADBCNN). In particular, the study aims to evaluate whether the performance gains arise from

1. the dual-branch design,
2. the cross-attention fusion mechanism,
3. their combined interaction.

This analysis is essential to demonstrate that the observed improvements are not incidental but are directly attributable to the proposed architectural choices.

6.2. Ablation components and variations

Four variants of the architectures have been assessed with the same training settings, preprocessing methods, and data splits:

1. **Single-Branch CNN (ResNet34):** The standard single-stream residual network was used as the baseline for the feature extraction process without the use of dual-branch interactions.

Table 3
Overall Accuracy Comparison of Proposed CADBCNN Model Performance with other architectures.

Model	Accuracy	Right (R)	Wrong (W)
ResNet34	0.9210	900	77
ResNet50	0.9620	940	37
EfficientNet-B0	0.9421	921	56
Dual-Branch (Feature Concatenation)	0.9564	934	43
Proposed CADBCNN	**0.9836**	961	16

Table 4
Comparison based on interpretability and limitations.

Model	Accuracy	Key characteristics	Limitations
Proposed CADBCNN	**0.9836**	Dual-branch architecture with cross-attention fusion; strong localization of disease-relevant regions and balanced class-wise performance	Lack of pixel-level localization metrics against expert annotations
ResNet34	0.9210	Deeper residual learning; robust baseline performance	No explicit attention or explainability
ResNet50	0.9620	Deeper residual architecture; improved feature abstraction	Increased computational complexity
EfficientNet-B0	0.9421	Efficient scaling of depth and width	Limited interpretability

Table 5
Architecture-level ablation results.

Model variant	Feature fusion strategy	Accuracy	F1-score
Single-Branch ResNet34	None	0.9210	0.9214
Dual-Branch (Feature Concatenation)	Static Concatenation	0.9564	0.9561
Dual-Branch (Additive Fusion)	Element-wise Addition	0.9612	0.9608
Proposed CADBCNN	Bidirectional Cross-Attention	**0.9836**	**0.9838**

2. **Dual-Branch Feature Concatenation:** This model consists of two parallel CNN streams with the extracted features combined via static concatenation without the use of adaptive interactions.
3. **Dual-Branch with Additive Fusion:** The second model with the features combined via additive fusion was also assessed to check if the performance can be enhanced through simple feature aggregation.
4. **Proposed CADBCNN (Dual-Branch + Cross-Attention):** This is the proposed model with the use of bidirectional cross-attention to allow dynamic and adaptive feature interactions between the two streams.

6.3. Quantitative ablation results

The quantitative results reveal the progressive improvement in the performance of the classifier with the increase in the complexity of the architectures and the ability to interact between the features. The performance of the baseline model with the single branch is the worst among the models. However, the performance improves with the addition of the second branch and the static concatenation of the features. The improvement in the performance is due to the ability to extract the features at multiple scales.

6.4. Architecture-level ablation results

Various model variants with different strategies for feature fusion are compared in Table 5 . It can be seen that using a dual branch structure can boost model performance compared to using a single branch model. Moreover, the proposed CADBCNN model using bidirectional cross-attention yields the best accuracy and F1 score.

6.5. Statistical significance analysis

For assessing the statistical significance of the performance gains of CADBCNN over the baseline architectures, McNemar test was applied to the paired test predictions. Furthermore, the 95% bootstrap confidence intervals were calculated for accuracy and F1-score metrics. The null hypothesis was set as having no significant difference between the models. The resulting p-values ($p < 0.05$) reveal that CADBCNN significantly outperforms the single-branch baselines, thus ensuring that the gains are not due to mere chance. Considering the stratified evaluation protocol and the test cohort size, the metrics provided can be said to be statistically sound and reflective of the generalization ability. Additionally, the qualitative feature visualization and Grad-CAM-based explainability methods can be said to provide interpretability to the model's predictions without making claims to clinical validation. Overall, the metrics provided can be said to effectively demonstrate the efficacy of the proposed CADBCNN model for the automated classification of leukemia stages.

6.6. Clinical relevance and taxonomic justification

The four-class system used in this study (Benign-like, Early, Pre, and Pro) follows the categories in the publicly available dataset. This dataset shows the morphological subtypes of acute lymphoblastic leukemia (ALL), including Early Pre-B, Pre-B, and Pro-B variants, along with benign hematogone cases. These categories do not indicate progressive clinical stages of a single type of leukemia. Instead, they represent distinct subtypes within the B-cell lineage. In standard hematopathology, a definitive leukemia diagnosis relies on combining peripheral smear morphology, flow cytometry immunophenotyping, cytogenetic analysis, and molecular testing according to World Health Organization (WHO) guidelines. The proposed model is not meant to replace these diagnostic standards. Instead, it serves as a decision-support tool for initial morphological screening. The finer details within B-cell precursor ALL may help with early morphological classification before laboratory confirmation. This could improve triage efficiency in settings with limited resources. The distinction between Early Pre-B, Pre-B, and Pro-B ALL highlights differences in immunophenotypic maturation stages rather than recognized independent disease types. Thus, the taxonomy used in this study should be seen as a morphological classification based on the dataset, not a substitute for WHO diagnostic guidelines.

6.7. Limitations and future scope

Although the proposed CADBCNN framework shows promising results, several limitations should be noted. First, the study uses a single publicly available dataset and lacks external multi-center validation. This may reduce the validity of the findings in different lab settings and staining methods. Second, the model does not include patient-level clinical data such as age, gender, immunophenotyping markers, and cytogenetic information. This lack of combining clinical information limits the framework to just image-based morphological analysis. Third, while Grad-CAM visualizations offer some interpretability, the assessment remains qualitative. It has not been validated against pathologist-annotated nuclear or cytoplasmic areas using quantitative localization metrics. Moreover, differences in microscope imaging conditions, staining strengths, and equipment could affect the model reliability when used in varied clinical environments. Future research should prioritize external validation using datasets from multiple institutions to test the findings across different acquisition settings. Combining clinical and lab data, including flow cytometry markers and cytogenetic profiles, could further improve diagnostic accuracy through multimodal learning. A quantitative assessment of saliency maps using expert-annotated segmentation masks and intersection-over-union metrics should be included to boost interpretability claims. Additionally, exploring transformer-based hybrid models or multi-scale attention methods could enhance feature representation and better capture complex cellular structures at different magnifications.

7. Conclusion

This work presents a Cross-Attentive Dual-Branch Convolutional Neural Network (CADBCNN) that explains the automated classification of leukemia white blood cell images into four categories: Benign-like, Early, Pre, and Pro. The framework focuses on reproducibility and fairness by using stratified data splitting, class-weighted loss optimization, and controlled data augmentation to reduce class imbalance and avoid data leakage. The main contribution of this study is the bidirectional cross-attention mechanism. This feature allows for information exchange between two parallel CNN branches, enabling the model to capture additional spatial and semantic features. Testing on an independent set shows strong generalization. The model achieved a test accuracy of 98.36%, a macro-averaged F1-score of 0.9838, and a macro-averaged AUC of 0.9995, supported by narrow 95% bootstrap confidence intervals. Comparisons with optimized CNN baselines confirm that the performance gains come from cross-attention-based feature fusion, not just from the network's depth. To improve transparency, Grad-CAM visualizations and t-SNE feature embedding analyses were added to evaluate model behavior qualitatively. These analyses indicate that the model pays attention to important diagnostic areas and maintains well-separated feature representations. However, it is important to note that interpretability is still supportive rather than clinically validated. Overall, CADBCNN is a dependable decision-support tool for preliminary morphological screening. It also lays the groundwork for future work on external validation, quantitative explanation, and the integration of multiple clinical modalities.

List of abbreviations

This study involves several commonly used abbreviations. AI refers to Artificial Intelligence, and DBCA stands for dual-branch cross-attention. WBC denotes white blood cells, while leukemia subtypes include acute lymphoblastic leukemia (ALL), acute myeloid leukemia (AML), and chronic myeloid leukemia (CML). XAI represents Explainable Artificial Intelligence techniques. The visualization tools included Gradient-weighted Class Activation Mapping (Grad-CAM) and t-distributed stochastic neighbor embedding (t-SNE).

CRediT authorship contribution statement

Krishna Kumar Jha: Writing – review & editing, Writing – original draft, Supervision, Software, Resources, Formal analysis, Data curation, Conceptualization. **Tridib Chakraborty:** Writing – original draft, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Sneha Saha:** Writing – original draft, Visualization, Software, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Sourya Pratim Das:** Writing – original draft, Visualization, Software, Resources, Investigation, Formal analysis, Data curation. **Tamannya Mukherjee:** Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Bappa Ditya Biswas:** Writing – review & editing, Writing – original draft, Visualization, Software, Conceptualization.

Ethical statement

This study does not involve human participants, human data, or animal subjects. The datasets used in this research are publicly available and were used strictly for research purposes. Therefore, ethical approval and informed consent were not required for this study.

AI Disclosure statement

During the preparation of this manuscript, the authors used OpenAI and PaperPal for language editing, grammar improvement. After its use, the author(s) thoroughly reviewed, verified, and revised all AI-assisted content to ensure accuracy and originality. The author(s) take full responsibility for the integrity and final content of the published article.

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Declaration of competing interest

All authors declare no conflicts of interest.

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Data availability

This study used publicly accessible datasets [33]. The datasets analyzed in the current study are available from the corresponding author upon request.

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